

Dilations: Building and Measuring Similar Figures

Answer Key

High School Geometry

Quick Answers

- | | | |
|-----------------------|-----------------|---------------|
| 1. 6, 3 | 4. 7, 11 | 7. 0.005, 0.4 |
| 2. 7.5, 12.5 | 5. 24, 36, 5.76 | 8. 6 |
| 3. $\frac{7}{2}$, 14 | 6. 6, 12, 2 | |
-

Curriculum

Level: High School Geometry

HSG-SRT.A.1 – problems 1, 2, 4

HSG-SRT.A.2 – problems 3, 5, 6, 7, 8

Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

7. If they got 0.06: divided the area by the scale factor instead of by its square

1. Projector: dilation centred at the origin, $k = 3$, applied to $A(2, 1)$ (centimetres).

Solution: With the centre at the origin, a dilation multiplies each coordinate by the scale factor, because A' must lie on ray OA at three times the distance: $(x, y) \mapsto (3x, 3y)$. So the first coordinate is $3 \cdot 2$ and the second is $3 \cdot 1$. $A'(6, 3)$

Check: $OA = \sqrt{2^2 + 1^2} = \sqrt{5}$ and $OA' = \sqrt{6^2 + 3^2} = \sqrt{45} = 3\sqrt{5}$, three times as far, on the same ray.

2. Sail: pattern $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ in metres, dilated about A with $k = 2.5$.

Solution: The centre is the origin, so multiply every coordinate by 2.5: $B' = (10, 0)$ and $C' = (10, 7.5)$. Side $B'C'$ runs straight up from $(10, 0)$ to $(10, 7.5)$, so its length is the difference of the y -coordinates, $7.5 - 0$ metres. 7.5

Side $A'C'$ runs from the centre $(0, 0)$ to $(10, 7.5)$: $\sqrt{10^2 + 7.5^2} = \sqrt{100 + 56.25} = \sqrt{156.25}$ metres. 12.5

Check the similarity claim: in the pattern $BC = 3$ and $AC = 5$, and $2.5 \times 3 = 7.5$, $2.5 \times 5 = 12.5$. Every length scaled by the same factor, which is what makes the sail similar to its pattern.

3. Blueprint: 6 cm by 4 cm enlarged so the long side becomes 21 cm.

Solution: The scale factor of a dilation is the image length divided by the matching pre-image length — always image over pre-image, never the other way round: $k = \frac{21}{6}$, which reduces. 7
2

Every length is multiplied by that same factor, so the enlarged width is $\frac{7}{2} \times 4 = 14$ centimetres. 14

Check: $6 \times 3.5 = 21$ ✓, and the enlarged rectangle is 21×14 , whose sides are still in the ratio 3 : 2.

4. Map zoom: centre $P(1, 2)$, $k = 3$, applied to $Q(3, 5)$ (kilometres).

Solution: The centre is not the origin, so measure from the centre, scale, and step back out: $Q' = (h + k(x - h), j + k(y - j))$ with $(h, j) = (1, 2)$.

Horizontally, Q sits $3 - 1 = 2$ km east of the pin; tripling gives 6 km east of the pin, so the image's first coordinate is $1 + 3(3 - 1)$. 7

Vertically, Q sits $5 - 2 = 3$ km north of the pin; tripling gives 9 km north, so the second coordinate is $2 + 3(5 - 2)$. 11

So $Q' = (7, 11)$. Note the pin itself does not move — the centre of a dilation is its only fixed point.

5. Photo: 10 cm by 15 cm enlarged with $k = 2.4$.

(a) Lengths are multiplied by the scale factor, so the two sides become 2.4×10 and 2.4×15 centimetres. 24

36

(b) Area is two-dimensional: both the length and the width are multiplied by 2.4, so the area is multiplied by $2.4 \times 2.4 = 2.4^2$. 5.76

Check with the actual areas: $10 \times 15 = 150$ and $24 \times 36 = 864$, and $864 \div 150 = 5.76$ ✓.

6. Triangle DEF: $D(-2, 1)$, $E(4, 1)$, $F(4, 5)$ in metres, dilated about the origin with $k = 2$.

(a) Multiply every coordinate by 2: $D'(-4, 2)$, $E'(8, 2)$, $F'(8, 10)$. Plotted, $D'E'F'$ is the same shape as DEF , twice as far from the origin in every direction.

(b) D and E share the y -coordinate 1, so DE is the difference of the x -coordinates: $4 - (-2)$ metres. 6

D' and E' share the y -coordinate 2, so $D'E' = 8 - (-4)$ metres. 12

(c) Dividing image length by pre-image length gives $\frac{12}{6}$, and that quotient is the scale factor itself. 2

That is the point of the problem: the ratio of any pair of corresponding lengths comes out to k , which is what "similar with scale factor k " means.

7. Model car: $k = \frac{1}{12}$; real length 4.8 m, windshield area 0.72 square metres; Ben says 0.06 square metres.

Solution: Ben scaled an *area* by the scale factor, computing $0.72 \div 12 = 0.06$. A dilation multiplies every length by k , so an area — a product of two lengths — is multiplied by k^2 . The model's windshield area is $0.72 \div 12^2 = 0.72 \div 144$ square metres. 0.005

The model's length is a length, so it does scale by k alone: $4.8 \div 12$ metres. 0.4

Sanity check on Ben's number: 0.06 square metres is a square about 24 centimetres on a side, which cannot fit on a car 0.4 metres long.

8. Triangle JKL : $J(1, 1)$, $K(3, 1)$, $L(3, 4)$, dilated about the origin with $k = 2$.

(a) Doubling each coordinate gives $J'(2, 2)$, $K'(6, 2)$, $L'(6, 8)$.

(b) K' and L' share the x -coordinate 6, so $K'L'$ is the difference of the y -coordinates, $8 - 2$. 6

Check: $KL = 4 - 1 = 3$ in the pre-image, and $2 \times 3 = 6$ ✓.

(c) (Open response — judge by reading.) Model answer: the three lines JJ' , KK' , LL' all pass through the origin — they meet at the centre of dilation. That is forced by the definition: each image point lies on the ray from the centre through its pre-image point.

Why the triangles are similar: every image length is exactly 2 times its pre-image length ($J'K' = 4 = 2 \cdot 2$, $K'L' = 6 = 2 \cdot 3$, $J'L' = 2\sqrt{13} = 2 \cdot \sqrt{13}$), so all three pairs of corresponding sides are in the same ratio 2 : 1, giving SSS similarity. Equivalently, a dilation is by definition a similarity transformation: it preserves angle measure and multiplies every length by k , so $\triangle JKL \sim \triangle J'K'L'$ with ratio 1 : 2.

Look for: the student states that the three lines are concurrent at the centre (the origin), and justifies similarity by a constant ratio of corresponding sides or by naming the dilation as a similarity transformation — not merely "they look the same shape".